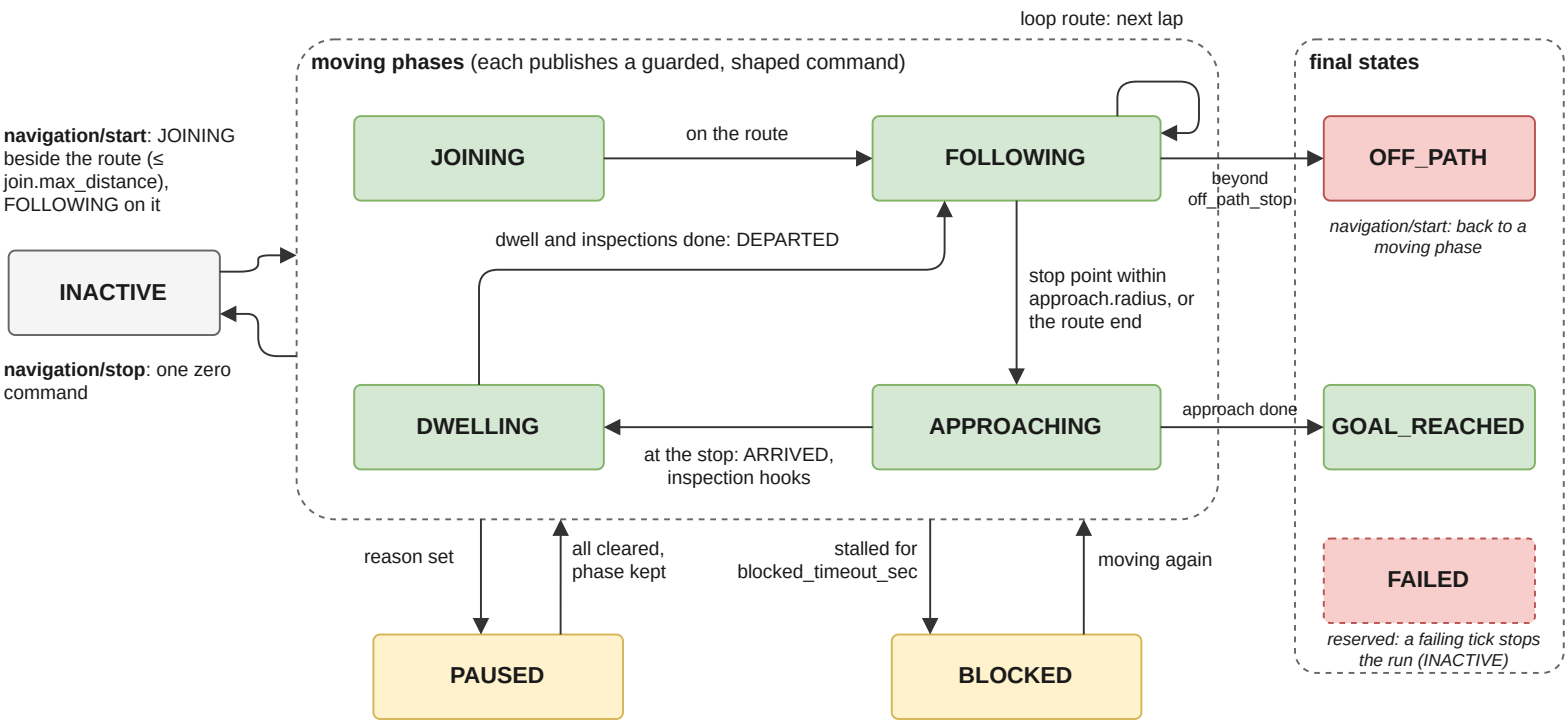

Pause reasons (a set): **operator** navigation/pause, cleared by resume or start · **estop** from control/status, sticky: cleared only by navigation/resume after the release · **remote** while the mux is in REMOTE · **localization** while LocalizationStatus is NONE · **battery** below critical_battery

Holds (zero every tick, phase kept, status message prefix): TF_STARTING, TF_STALE (pose predicted with the odometry twist between tf_compensate_sec and tf_timeout_sec), TF_CLOCK, SCAN_STALE, SCAN_BLIND, AVOIDANCE_STALE, EMERGENCY; the last two and a stalled guard count toward BLOCKED